Explanations Using Defeasible Rule Sets

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Where do AI models explanations come from?

Classifiers make **high-stakes** calls: loans, diagnoses.

People need to know **why**, but most models are black boxes.

Rule-based classifiers learn **IF-THEN rules** from data, so every prediction comes with the rule that made it.

The problem

Real data has exceptions, so rules **conflict**. ECCRS settles it with a procedure: the most specific rule wins.

QUESTION

Is that sound reasoning, or just a convenient tie-breaker?

Defeasible reasoning is the formal logic of exceptions. Madzime & Meyer proved the two agree under **three assumptions**. We built a workbench to test it on real rule sets.

ECCRS: the explanation is the computation

MOST SPECIFIC WINS

Applicable rules

$$\mathcal{A}(F) = \{(B, \ell) \in \mathcal{R} : F \vDash B\}$$

Inclusion-maximal rules

$$\mathcal{M}(F) = \{(B, \ell) \in \mathcal{A}(F) : \nexists (C, \ell') \in \mathcal{A}(F) \text{ with } B \subset C\}$$

MSW operator

$$\mathrm{MSW}_{\mathcal{R}}(F) = \begin{cases} \ell & \text{if all rules in } \mathcal{M}(F) \text{ have label } \ell \\ \bot & \text{if } \mathcal{M}(F) = \varnothing \quad \text{(abstain)} \\ \text{halt} & \text{if } \mathcal{M}(F) \text{ has both labels} \end{cases}$$

EXPLANATION TRACE

r1: odour = none ⇒ 1 (edible)

r2: odour = none ∧ spore = green ⇒ 0

r3: odour = foul ⇒ 0

F: odour = none, spore = green

$\mathcal{A}(F) = \{r_1, r_2\}$

$B_{r_1} \subset B_{r_2}$, so r1 is overridden

$\mathcal{M}(F) = \{r_2\}$

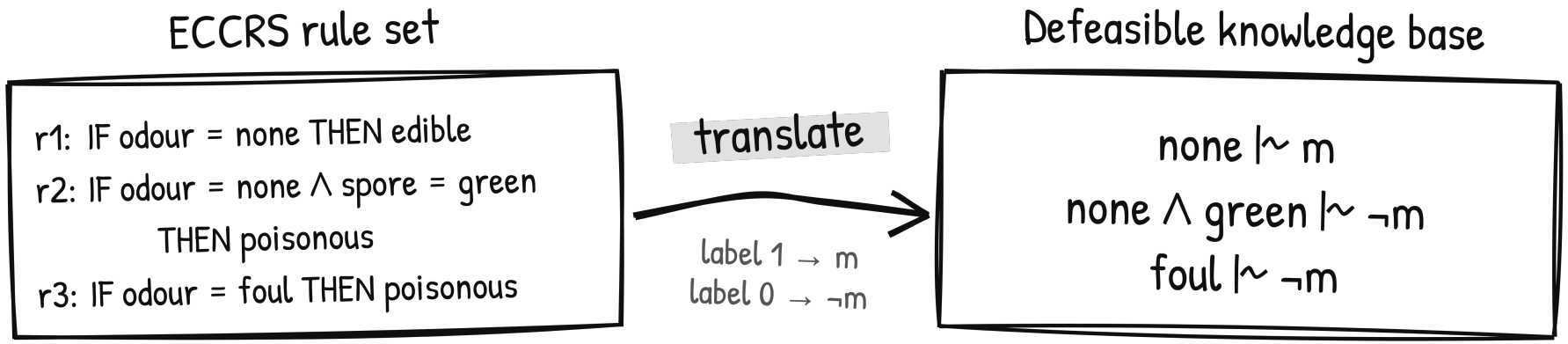
$\mathrm{MSW}(F) = 0$ **poisonous**

*The steps that compute the answer **are** the explanation.*

From rules to defeasible reasoning

DEFEASIBLE REASONING (KLM)

$\text{bird} \mathrel{|\!\sim} \text{flies}$ $\quad \text{penguin} \rightarrow \text{bird}$ $\quad \text{penguin} \mathrel{|\!\sim} \neg\ \text{flies}$ the specific statement wins



Three closures, one question

RATIONAL CLOSURE

Discards whole ranks of statements, most general first.

Daniel Schlesinger

LEXICOGRAPHIC CLOSURE

Keeps as many statements from the clashing rank as it can.

Augustine Mochoeneng

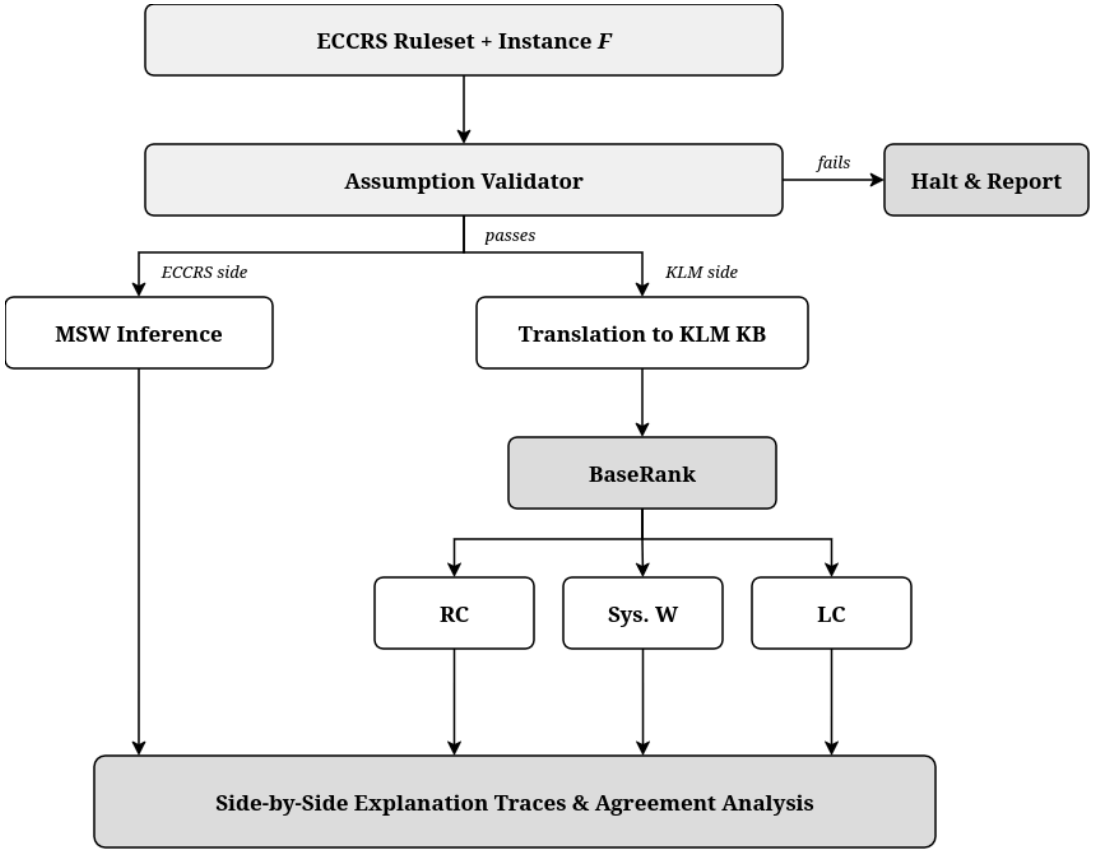
SYSTEM W

Compares worlds by the set of statements they break.

Antony Baker

For every instance F: does $F \mid\!\sim m$ or $F \mid\!\sim \neg m$?

The workbench



Results

ASSUMPTIONS HOLD

Instances	RC	LC	System W
13,509	**100%**	**100%**	**100%**

Bank, Breast Cancer and Mushroom rule sets.

ASSUMPTIONS BROKEN (MSW VS LC)

Broken	Agree	Abstain	Conflict	Wrong
Closure	13,395	114	0	**0**
Sanity	12,019	0	1,490	**0**
Total override	13,374	0	135	**0**

13,509 instances per row.

CONCLUSION

ECCRS explanations are not just readable. They are backed by formal reasoning.